

Profile Card Explorer

JavaScript DOM Mini Project | 1st Year BTech CSE | Time: 20-25 minutes

Topics used: document object, getElementById, getElementsByClassName, getElementsByTagName, querySelector, console.log

You get two ready-made files: `index.html` (the page, already written, do not edit it) and `script.js` (empty, you write your JavaScript here). Open `index.html` in Chrome, press **F12**, and look at the **Console** tab to see your output.

The Story

A classmate has built a small **Student Profile** page, but it is boring and nobody can change it. You are the JavaScript detective. Your job is to **find** different parts of the page using JavaScript, **read** what is inside them, and **change** a few of them. Think of the page as a city and the selecting methods as different ways of asking for an address.

Quick Recall

- **Document** = your whole web page, loaded into JavaScript. Everything on the page lives inside it.
- **Object** = a thing that has **properties** (information, like `document.title`) and **methods** (actions, like `document.getElementById()`).
- The **dot** means "go inside": `document.title` reads as "from the document, give me the title".
- A selecting method gives back **one element** or a **collection** (many elements, like a list that starts counting from 0).

Your Toolbox

`element.textContent` gives the text inside an element, and you can also assign new text to it.
`element.style.color` changes the text colour, for example = "red".
`collection.length` tells you how many elements are inside a collection.
`collection[0]` is the first element, `collection[1]` is the second, and so on.

Your Tasks

Task 1: Meet the document

Print three things in the console: the document itself, the **title** of the page, and the **type** of document (use `typeof`).

***Hint:** The title is a property of the document object. Use the dot to go inside. Your `typeof` answer should explain why we call it the "document **object**".*

Task 2: Find by ID

The big heading at the top has an **id**. Select it using `getElementById` and store it in a variable. Print it, then **change its text** to your own name, like "Aman's Profile".

***Hint:** Open `index.html` and look at the heading's id. Pass that exact text to the method, inside quotes. Should the variable be `const` or `let`? You are not changing which element it points to.*

Task 3: Find by class name

Three lines of the profile (branch, year and city) share the same **class**. Select all of them using `getElementsByClassName`. Print **how many** you found, then print the **text of the second one**.

*Hint: This method returns a collection, not a single element, so one element needs a position number. Programmers count from 0, so which number means "second"? Notice the name has an **s** in Elements.*

Task 4: Find by tag name

Select all the list items (`li`) on the page using `getElementsByTagName`. Print how many there are and print the text of the **first** one.

Hint: For tag names you pass just the name of the tag, without the angle brackets. The result is a collection again, so think back to Task 3.

Task 5: The all-rounder, `querySelector`

Now do the job using `querySelector`, which uses the same selectors you wrote in CSS. Select these **three** elements with it:

- The paragraph with id **name**; print its text
- The **first** element with class **skill**; print its text
- The element with class **hot**; print its text and change its colour to any colour you like

*Hint: Recall from CSS: an **id** starts with one symbol and a **class** starts with another. Write them exactly as you did in a stylesheet, inside quotes. `querySelector` returns only one element, so no position number is needed.*

What your console should show (roughly)

```
Page title: Profile Card
Type of document: object
Details found: 3
Second detail: Year: 1st
List items found: 3
First list item: HTML
Name: Name: Rahul
First skill: HTML
Hot skill: JavaScript
```

Your wording may differ. What matters is that the values are correct. In the browser, the heading should show your name and the JavaScript skill should have your chosen colour.

Task 6: Break it on purpose!

Select an id that **does not exist**, for example `getElementById("nme")` (a typo). Print the result and see what you get. Then try to **change its textContent** and read the red error message in the console.

In a comment (a line starting with `//`), write **one sentence** explaining what happened.

Hint: When JavaScript cannot find anything, it gives back a special "nothing" value. The error is about trying to use a property on something that is not there. Also: which of your tools does a typo in the id break?

Bonus Challenges (for fast finishers)

- ★ **Bonus 1:** Use `querySelectorAll` to select **every** element with class `skill`. Print how many you found and the text of the last one.
- ★ **Bonus 2:** Select the paragraph **inside** the footer div using a single `querySelector` call, and change its text.

Hint: For Bonus 1, the name hints at the answer (select... all). It returns a collection, so use position numbers again; the last position is one less than the length. For Bonus 2, CSS lets you write a parent and a child separated by a space.

Fill This Table at the End

Based on what you saw in your console, write what each method returns: **one element** or a **collection**?

Method	What you pass in	Returns?
<code>getElementById</code>	an id name	
<code>getElementsByClassName</code>	a class name	
<code>getElementsByTagName</code>	a tag name	
<code>querySelector</code>	a CSS selector	

Before You Submit: Self-Check

- ☐ I printed the document, its title and its type in Task 1
- ☐ I used all three `getElementBy...` methods and one `querySelector`
- ☐ I used position numbers correctly for collections (counting starts from 0)
- ☐ The heading shows my name and one skill has a new colour
- ☐ I broke it on purpose, read the error and wrote my explanation as a comment

Stuck? Check for typos in names first, then print the variable with `console.log` to see what you really have. Most DOM bugs are just a wrong name or a missing position number.